Similarity Measure

	T1	T2	T3	T4	T5	T6	T7	T8
DOC1	0	4	0	0	0	2	1	3
DOC2	3	1	4	3	1	2	0	1
DOC3	3	0	0	0	3	0	3	0
DOC4	0	1	0	3	0	0	2	0
DOC5	2	2	2	3	1	4	0	2

Find similarity index using DOC2, DOC4 for

i) Cosine Similarity Measure

~~ii) Jaccard Similarity~~

~~iii) Pearson's correlation~~

Cosine Similarity Measure

$$\text{Dot_Product}(DOC2, DOC4) = \langle 3,1,4,3,1,2,0,1 \rangle * \langle 0,1,0,3,0,0,2,0 \rangle$$

$$= 3*0 + 1*1 + 4*0 + 3*3 + 1*0 + 2*0 + 0*2 + 1*0 = 10$$

$$\text{Norm}(DOC2) = \sqrt{3^2+1^2+4^2+3^2+1^2+2^2+0^2+1^2} = 6.4$$

$$\text{Norm}(DOC4) = \sqrt{0^2+1^2+0^2+3^2+0^2+0^2+2^2+0^2} = 3.74$$

$$\text{Cosine}(DOC2, DOC4) = \frac{\text{Dot_Product}(DOC2, DOC4)}{\text{Norm}(DOC2) * \text{Norm}(DOC4)}$$

$$= \frac{10}{6.4 * 3.74} = 0.42$$

So, DOC2 and DOC4 are 42% similar

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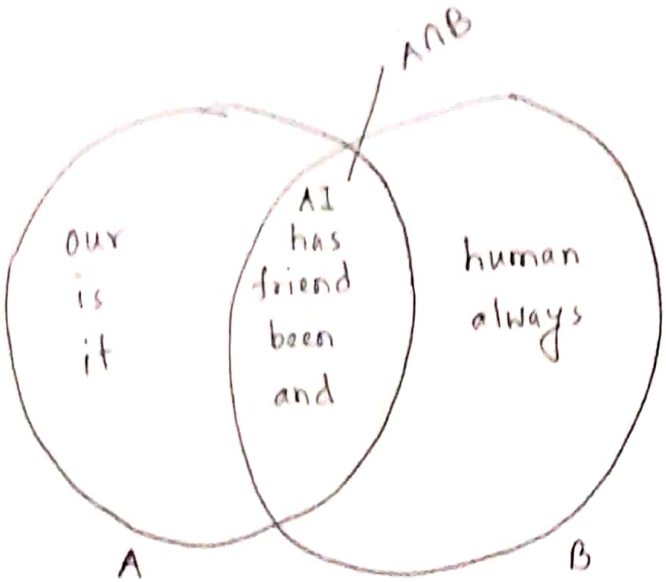
Jaccard Similarity

Ques: Sentence-I : AI is our friend and it has been friendly

Sentence-II : AI and humans have always been friendly

Find Jaccard similarity Index for sentence-I and sentence-II

Ans:



Lemmatization: (Reduce words to the same root word)

Here, friend, friendly → friend

has, have → has

$$J(\text{Sentence 1, Sentence 2}) = \frac{|A \cap B|}{|A \cup B|}$$

$$= \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

$$= \frac{5}{8+7-5}$$

$$= \frac{5}{10}$$

$$= 0.5$$

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